

EDUCATION

Purdue University, USA

Master of Science, Mechanical Engineering (minor: Computational Science)
GPA: 3.67/4 | Aug 2014 – Dec 2016

Vellore Institute of Technology, India

Bachelor of Technology, Mechanical Engineering
GPA: 9.03/10 | Aug 2010 – Jun 2014

SKILLS

Thermal Engineering

- Thermal validation & testing
- Thermal architecture & design
- IR imaging & thermocouples
- Software thermal management policies
- Failure analysis

Simulation & CAD

- Ansys Icepak
- FloTherm XT
- COMSOL
- SolidWorks CAD

Product Development & Programming

- Consumer electronics development (xVT)
- Test automation frameworks
- Python & Unix scripting
- MATLAB & data analysis

PUBLICATIONS

Experimental investigation of Phase Change Materials for thermal management of handheld devices

International Journal of Thermal Sciences (2018)

Investigation of aluminum foams and graphite fillers for improving the thermal conductivity of paraffin wax-based phase change materials

IEEE ITherm (2017)

Passive Thermal Management Using Phase Change Materials: Experimental Evaluation of Thermal Resistances

ASME InterPACK (2015)

LANGUAGES

English (Professional), Mandarin Chinese (HSK Level 2)

SUMMARY

Thermal engineer who ships consumer electronics. 8+ years taking products from first concept through mass production — thermal architecture, thermal validation (IR imaging, thermocouples, automated test frameworks), and software thermal management policies and test methods that scale across product lines.

PROFESSIONAL EXPERIENCE

Apple Aug 2023 – Present

Thermal Engineer (System Product Design)

Thermal testing and design of iPad Air (6th & 7th gen, 2024/2025) and iPhone 17 (2025).

- Own all thermal objectives from initial concept through launch, spanning thermal architecture, measurement and software thermal management
- Analyze field metrics to understand real-world customer usage patterns and develop representative thermal tests
- Set up a thermal validation lab at a vendor site and defined thermal test SOPs
- Develop frameworks to continually optimize thermal testing and data analysis
- Communicate thermal performance and risk to cross-functional teams, product managers and executives

Amazon Lab126 Dec 2018 – Jul 2023

Thermal Engineer II

Thermal testing and design of Echo Dot (3rd & 5th gen), Echo (3rd gen), Echo Show 10, Echo Studio, Echo Flex, Halo Rise and the Astro home robot.

- Enabled \$250k cost savings for Echo Dot by validating and assessing thermal risk during new product development
- Led thermal validation of Amazon's first consumer robot (Astro), ensuring thermal health and safety
- Built a Python test-automation framework that cut test time by 50% by automating music/video streaming, display and LED control, and internal temperature and power capture
- Developed the Temperature Sensor feature on Echo Dot (5th gen) by collecting and processing large-scale device data with Python and Unix scripting

Syscon International Feb 2017 – Oct 2018

Product Engineer

Developed portable temperature measurement systems from concept to manufacturing for the Iron & Steel industry.

- Standardized thermocouple calibration; tested and troubleshoot hardware and software prototypes
- Sped up product builds by 25% with MATLAB algorithms for processing field data
- Trained distributors and sales teams; presented at AFS and AIST industry conferences

Purdue University Jan 2015 – Dec 2016

Graduate Research Assistant

Master's thesis: cooling smartphones using Phase Change Materials.

- Developed a passive cooling solution for consumer electronics, cutting processor temperatures by 5%
- Performed thermal design and CFD analysis in COMSOL and Ansys Fluent; validated with IR camera and thermocouple measurements
- Published in the International Journal of Thermal Sciences; presented at ASME InterPACK